Wenbo Deng

PhD Position

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The Chinese University of Hong Kong, Shenzhen   Computer and Information Engineering • M.Phil.
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Master's student in **Computer Science** at **The Chinese University of Hong Kong, Shenzhen** (specializing in LLM reasoning and training) with a Physics background **(GPA 3.81/4.3, Ranked 3/67, Top 5%)**. Possesses strong mathematical logic and algorithm modeling capabilities. Independently led **Large Language Model reasoning** research during Master's studies, with **G^{2}RPO-A accepted to ACL 2026 Main Conference (CCF-A) as co-first author**. Has solid PyTorch engineering implementation skills and top-tier research experience, capable of quickly undertaking cutting-edge algorithm exploration and reproduction tasks.

Research Interests

- LLM Reasoning; Multimodal Large Language Models (MLLMs); Audio Agents; Vision-Language Navigation (VLN) for embodied UAV navigation

Education

2024.09 2020.09	School of Physics, **Hefei University of Technology (211 Project University)** Bachelor of Science in Applied Physics **GPA 3.81/4.3, Ranked 3/67, Top 5%**
Present 2024.09	School of Science and Engineering, **The Chinese University of Hong Kong, Shenzhen** Master of Philosophy in Computer and Information Engineering

Research Publication

- Yongxin Guo, **Wenbo Deng** (Co-first Author), Zhenglin Cheng, and Xiaoying Tang. **G^{2}RPO-A: Guided Group Relative Policy Optimization with Adaptive Guidance**. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 Main Conference, CCF-A)*, pages 4525–4539. G^{2}RPO-A

Computer Skills

- Proficient in Python and mainstream deep learning frameworks, experienced in large model deployment, fine-tuning, and API integration
- Experienced in managing compute cluster resources using Slurm and K8s+Docker for training job creation and submission
- Developed synthetic data generation pipelines based on SOTA model distillation, combined with multi-granularity deduplication and cleaning strategies to build high-quality corpora for large model alignment

Internships

Now April 2026	RL Research Scientist (Intern) @ Autel Technology Inc.

VLM-Based Parking-Violation Annotation & RL Training Data Pipeline

- **Per-Box VLM Annotation Pipeline:** Designed a per-box (rather than per-frame) labeling prompt scheme for VLM-based parking-violation annotation, eliminating mixed multi-violation cases by construction; engineered structured JSONL output formatting and validated parsing reliability.
- **RL-Zero Training Data Strategy:** Launched RL-Zero training after finding rule-based judging accuracy sufficiently high while SFT data construction was costly; designed a curriculum that first trains output-format compliance without ground truth before introducing correctness rewards.

- **Evaluation Dataset & Methodology:** Built the test dataset and optimized violation-judgment F1 to 0.89; diagnosed evaluation validity issues including extreme class imbalance, semantic mismatch between frame-level labels and box-level scoring, and coupled highway/emergency-lane judgments; drove scenario-distribution expansion accordingly.
- **Tooling:** Developed a label-viewer tool for inspecting per-frame images against per-box annotations, accelerating annotation QC and error analysis.

VLM-Based Closed-Loop Autonomous Drone Flight-Control System

- **Training-Free VLM Flight Control:** Developed a training-free Vision-Language Model flight-control framework. The VLM performs high-level task reasoning and skill selection, while a hierarchical skill system handles low-level waypoint navigation, velocity control, yaw alignment, visual servoing, and safe landing.
- **Deployment Scenarios:** Implemented an autonomous vehicle-inspection workflow: detect a target vehicle, infer the nearest photographable vehicle end, navigate to a safe plate-capture station, perform 10× optical zoom evidence capture, and return to the original patrol waypoint.

Projects

- G^2RPO-A [Co-first Author; **Accepted to ACL 2026 Main Conference (CCF-A)**]: (Python, Shell, Linux)
  - **Problem & Solution**: Investigated the "Advantage Vanishing" problem caused by reward sparsity in GRPO-based Reinforcement Learning for Small-sized Large Language Models (SLMs), where high generation failure rates yield an extremely sparse reward matrix that blocks advantage-signal extraction. Proposed a CoT Prefix-based Guidance mechanism with an **Adaptive Partial Guidance** strategy – combining Curriculum Learning principles, dynamically adjusting guidance intensity and sample difficulty based on real-time training state – maintaining effective advantage signals while ensuring reward density.
  - **Theoretical Investigation**: Revealed that Full-Guidance (all completions in one group guided) causes reward saturation, driving the advantage-function variance to zero and eliminating update gradients; verified that efficient GRPO training hinges on constructing effective update signals from moderately difficult samples, rather than simply reducing task difficulty.
  - **Algorithm Performance**: Consistently and significantly outperforms GRPO, Clip-Higher, and SFT baselines across model sizes (0.6B, 1.7B, 8B) on mathematical reasoning (MATH500, AIME, GPQA) and code generation (HumanEval, LiveCodeBench); e.g., on Qwen3-1.7B, MATH500 accuracy improved from 79.49% to 82.08%, and AIME24 from 56.67% to 63.33%.
- `Mixture of Attention`: (C++, Python, Shell, Linux)
  - **Interpretability Analysis**: Applied the Mixture of Experts framework to Transformer multi-head attention mechanisms. Conducted panoramic deconstruction of response patterns across 144 ViT attention heads and dataset-level statistics over 1.44×10^5 attention matrices (1,000 samples), identifying functionally differentiated heads – "Local Heads" in shallow layers, "Global Heads" in deep layers, and silent background-only heads – whose activation patterns remain distributionally consistent at scale, highly consistent with SOTA Transformer-interpretability findings.
  - **Hardware-Software Co-design**: Constructed attention saliency masks (Saliency Mask/Topology) from the analysis and developed custom CUDA attention kernels aligned with mask characteristics and GPU architecture (thread-block mapping, memory coalescing), skipping invalid computation regions; reduced time complexity from $\mathcal{O}(3200d)$ to $\mathcal{O}(120d)$ and significantly improved attention-computation parallel efficiency and throughput.